Vadim Uvarov

ML Engineer (production & MLOps) | Data Scientist

PhD in Machine Learning

Industrial experience: 8 years

Location: Antwerpen, Belgium

Open to work: on-site/hybrid in Belgium; remote abroad

ML Engineer with 8 years of experience in production ML, MLOps, and data science across energy, telecom, and AI startup environments. Delivered cloud, on-prem and on-device ML solutions and ETL pipelines, including deployment, monitoring and stakeholder communication.

Having experience both in academy and industry, I can formalize a loosely stated business task, gather inputs from stakeholders and conduct a research & development of a solution either independently or as a part of a team.

I also have an experience of leading a data science team specialized in telecom geodata analysis.

Contacts

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Links: LinkedIn; My Website; Telegram;

Languages

English / C1 (advanced)

Dutch / A2 (elementary and improving)

Russian / C2 (native)

Visa Status

I'm eligible to work as a freelancer in Belgium (or abroad) and don't require a visa sponsorship.

Skills

Machine Learning & Deep Learning

- **Classic ML** (classification, regression, clustering, anomaly detection)
- **Time-series forecasting** (energy demand forecasting)
- **Deep learning** (TensorFlow, PyTorch)

AI Engineering (Gen AI)

- **LLMs** & **RAG** (Amazon Bedrock)
- Familiar with **LLM**-based applications and actively expanding expertise in this area

Data analysis & visualization

- Statistics / Hypothesis testing
- Dashboards (**Streamlit**, **simple frontend**)
- Plots: **matplotlib**, **seaborn**, **plotly**

Python (advanced)

- **Pandas/Polars**, **Scikit-Learn** for data science
- **numpy** / **scipy** for algo implementation
- **pytest** for unit testing
- **commitizen, ruff, mypy** for clean code

Cloud Deployment

- **AWS** (primary), **Kubernetes** (K8s), **Docker**
- **CI/CD:** Gitlab CI/CD, GitHub Actions
- Infrastructure-as-a-code: **Terraform**, **Pulumi**

MLOps & Production ML

- End-to-end ML lifecycle
- Cloud ML development and deployment (**Amazon Sagemaker**)
- Experiment tracking (**MLflow**)
- Observability: **Grafana**, **Prometheus**
- Model performance tracking, drift detection and retraining strategies

Big Data

- **Postgres, Teradata, Hadoop**, **Spark**
- **SQL**: joins, aggregates, window functions

Software Engineering

- **Git**, unit and integration **testing**, code review, style guide, production-ready code,
- AI-assisted (agentic) coding (**Claude Code CLI, GitHub Copilot**)

Work Experience

ML engineer – Centrica [energy company] – Antwerp, Belgium
Apr 2023 – Oct 2025

Centrica Business Solutions helps organizations to develop smart energy solutions.
I worked as a quantitative developer (a blend of a software engineer, data scientist and ML engineer).

- Designed and deployed an **electric vehicle (EV) charging controller**, enabling thousands of UK residents to save on their electricity bills, making it the cheapest way of EV charging in the UK.
- Built production-grade residential **energy demand forecaster** increasing savings by at least 15%.
- Created and deployed a physical-based **solar energy forecasting** model that significantly improved accuracy versus the baseline while preserving interpretability.

My tech stack here was: python, AWS, sklearn, pytorch, Terraform/Pulumi, K8s, Sagemaker, Grafana.

Data Scientist – Sentiance [AI startup] – Antwerp, Belgium
May 2021 – Jan 2023

Sentiance develops a Mobile SDK which provides insights into users' motion data. I worked on the lifestyle component. It was decided to migrate the solution from backend to the on-device.

- developed a **venue type mapping** model with **Python** and **TensorFlow** for on-device usage to predict venue type visited by user (e.g. shop/restaurant/bar, etc.)
- applied **transfer learning** to improve **venue type mapping** model accuracy
- tracked model training in **MLflow** to support reproducible experiments and ensure quality
- used **TensorFlow (lite)** to develop a multiplatform **on-device** implementation of the algorithm

This allowed for a blazingly fast on-device inference which also increased users' privacy.

Data Scientist -> Senior Data Scientist – Tele2 Russia [telecom] – Moscow, Russia
Jun 2018 – Apr 2021

At Tele2 telecom company I worked (first as individual contributor and then as a lead) of geodata team.

I've built and deployed several **big data** advanced analytics **ETL pipelines** with using **Spark** and **Airflow**:

- **track processing** for subscriber geolocation using triangulation technique (enhanced accuracy)
- migrated and enhanced **point-of-interest** detection (home/work) pipeline from **Teradata** to **Spark**
- **map matching** solution in **Scala Spark** to snap millions of subscribers' geolocation to a particular street/road segment (here I applied findings from my PhD thesis).

All that enhanced accuracy, reduced time-to market and provided ML features for the other teams.

Data Scientist – Magnit [food retailer] – Krasnodar, Russia
May 2017 – May 2018

I worked on a **retail site selection** problem: predict the shop potential revenue based on its coordinates.
I was training the model, designing the features and supervised the implementation in production.

Another project was **on-shelf availability** prediction system (based on time-series forecasting, not computer vision) to detect and promptly signalize that a particular product should be restocked.

Education

Novosibirsk State Technical University – PhD in Machine Learning
2015-2019

My research was focused on hidden Markov models. I have defended my PhD thesis in February 2020.

Novosibirsk State Technical University – MSc in Applied Mathematics and Computer Science
2010-2015

The subject of my graduate thesis was related to parallel programming on GPU (OpenCL).

My projects, articles, speeches

Pet project: event-driven task queue on AWS

Built an event-driven task queue on AWS (Lambda, SQS, Terraform) to deepen my cloud expertise. Repo: https://github.com/vadim-uvarov/aws-demo-simple-task-queue.

Research papers

Recent paper: "On-Device Deep Learning Location Category Inference Model" by Gadzhi Musaev, Kevin Mets, Rokas Tamosiunas, **Vadim Uvarov**, Tom De Schepper, Peter Hellinckx (Belgium, 2022)

My Research Gate profile: https://www.researchgate.net/profile/Vadim_Uvarov

Medium article "Feature selection using hypothesis testing"

https://medium.com/@vadim_uvarov/feature-selection-using-statistical-testing-2d8e7b5e27b8

Work Project "Map Matching of Cellular Data using hidden Markov models"

My conference speech about the technical aspects of this project (in Russian with English subtitles).

Scientific paper about this project, written in collaboration with my colleagues (in Russian, page 32).

PhD thesis "Incomplete sequence analysis using hidden Markov models"

Thesis (in Russian): www.nstu.ru/files/dissertations/dissertaciya_1574424031_157443103786.pdf

Video of my thesis defense (in Russian): link. Code and experiments: link.

Pet project: street signs character recognition (computer vision, deep learning)

I decided to implement a convolutional neural network for classifying images of letters (capital and lowercase) and digits in the wild (Chars74k dataset). I used Python, Theano and Lasagne library for CNN construction, Keras for data augmentation and OpenCV for image preprocessing (code on GitHub). I achieved 83% accuracy score and 14-th place in Kaggle training competition.

Recent Courses taken and Books read

Udemy course: AWS Certified Machine Learning Engineer Associate MLA-c01

Udemy course: AWS Certified Developer Associate 2025 DVA-C02.

Book: "Designing Data-Intensive Applications" by Martin Klepmann.

Book: "Designing Machine Learning Systems" by Chip Huyen.

Coursera specialization (watched lectures): Deep Learning by Deeplearning.AI.

Coursera specialization: machine learning and data science by Yandex.

Course (watched lectures): CS231n CNNs for Visual Recognition by Andrej Karpathy.

I've taken Dutch courses, which allowed me to achieve a certified A2 level in Dutch. I'm reading books and practice conversations in Dutch and to further advance my language skills.

Hobbies

To let my mind unwind after work, I play padel, go to gym and explore Belgium. I play electric guitar and enjoy video games. I like travelling and getting familiar with foreign cultures. I'm an AI enthusiast and use Perplexity for my daily life as well as agentic coding tools (Claude Code CLI) for side projects.